ECE569 — Homework 4

Histogram Generation on GPU

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Introduction

In this lab we set up to compare different implementations of histogram generation on GPU. The histogram is a fundamental data structure that counts the frequency of appearances of a certain data value in a dataset, commonly used to estimate the probability distribution of such data. Histograms find applications in computer vision, data analytics and scientific computing. The fundamental challenge of histogram generation on GPU is that when multiple threads attempt to update the same memory location (bin) a memory conflict may occur. Avoiding this conflicts are the main optimization ideas we will explore in this lab, followed by efficient data communication and probably efficient encoding.

In this Lab we have coded multiple implementations of histogram generation, from a global memory version with atomic operations (V0), to a shared memory block-privatized version (V1), to a vectorized $\mathcal{R}$=2-per-block version with RLE compression (V2, the GOAT), to an ablation of V2 without RLE (V3), to a warp-aggregated version (V4), to a bin-centric gather version with zero atomics (V5), to a sort-and-reduce approach via Thrust (V6), and finally to a multi-split hierarchical bucketing version (V7).

Phase 1: Hypotheses & Mental Models

Before running a single line of code, an optimization engineer builds a mental model of what *should* happen. Read through these core concepts and form your hypotheses.

1 Experiment 1: Random Data (Dataset 6)

Dataset 6 contains 500,000 elements drawn uniformly at random from [0, 4095], producing a nearly flat histogram across all 4096 bins. Each kernel version was executed 10 times; kernel execution time (excluding host–device data transfer) is recorded in milliseconds.

2 Experiment 2: Uniform Data (Same Value)

The uniform dataset contains 500,000 elements all set to the same value (42), directing all atomic traffic to a single bin. This stress-tests contention in each implementation.

3 Analysis 1: V0 vs. V1 — Global vs. Shared Memory

V0 vs. V1 on Experiment 1 (random data)

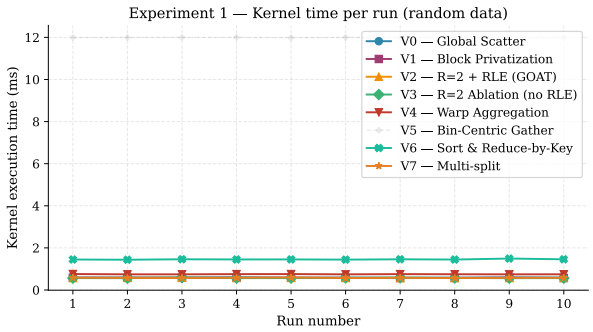
[TODO: Compare 0.5929 ms (V0) vs. 0.5774 ms (V1). Privatization moves atomic operations from global DRAM to on-chip shared memory. With random data, 500k elements spread across 4096

Table 1: Kernel execution time — Experiment 1 (random, 500k elements, 4096 bins, clipped at 127)

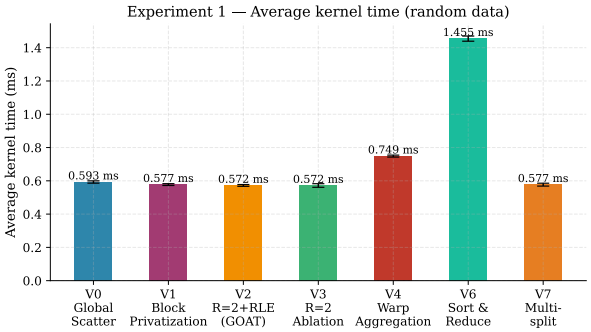
Run	V0 (ms)	V1 (ms)	V2 (ms)	V3 (ms)
1	0.5900	0.5847	0.5663	0.5663
2	0.5974	0.5755	0.5801	0.5621
3	0.5926	0.5790	0.5808	0.5912
4	0.6074	0.5769	0.5739	0.5648
5	0.5886	0.5810	0.5739	0.5664
6	0.5877	0.5774	0.5734	0.5773
7	0.5845	0.5709	0.5647	0.5893
8	0.5873	0.5732	0.5698	0.5735
9	0.5990	0.5862	0.5677	0.5618
10	0.5939	0.5697	0.5728	0.5662
Avg	**0.5929**	**0.5774**	**0.5723**	**0.5719**
Stdev	**0.0069**	**0.0055**	**0.0054**	**0.0108**

Table 2: Kernel execution time — Experiment 1, V4–V7 (random, 500k elements, 4096 bins, clipped at 127)

Run	V4 (ms)	V5 (ms)	V6 (ms)	V7 (ms)
1	0.7579	11.9899	1.4472	0.5847
2	0.7434	11.9961	1.4374	0.5725
3	0.7435	11.9959	1.4594	0.5688
4	0.7532	11.9982	1.4508	0.5809
5	0.7556	11.9826	1.4525	0.5940
6	0.7430	11.9950	1.4426	0.5671
7	0.7537	11.9871	1.4578	0.5815
8	0.7472	11.9991	1.4458	0.5667
9	0.7434	11.9848	1.4939	0.5761
10	0.7452	11.9822	1.4577	0.5781
Avg	**0.7486**	**11.9911**	**1.4545**	**0.5770**
Stdev	**0.0058**	**0.0065**	**0.0156**	**0.0087**



(a) Run-by-run timing



(b) Average with std dev

Figure 1: Experiment 1 — kernel execution time, random data

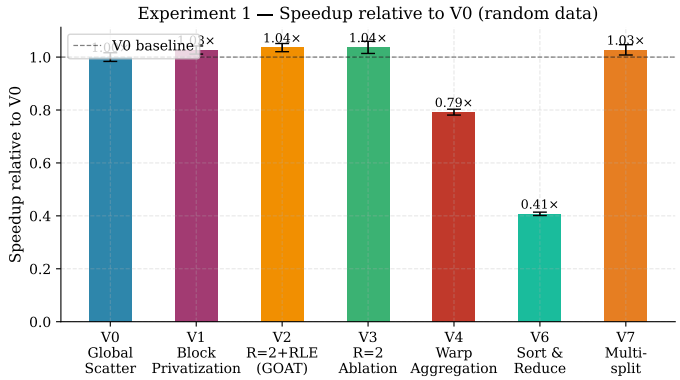


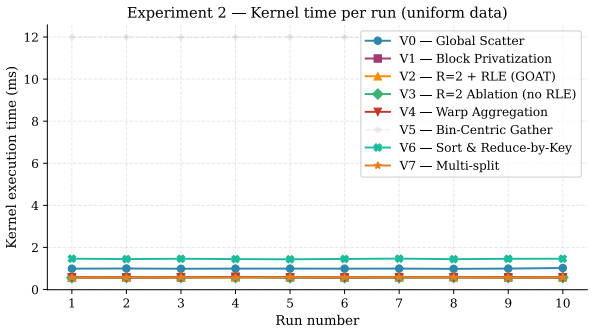
Figure 2: Experiment 1 — speedup of V1, V2, V3 relative to V0 baseline

Table 3: Kernel execution time — Experiment 2 (uniform, 500k elements, 4096 bins, clipped at 127)

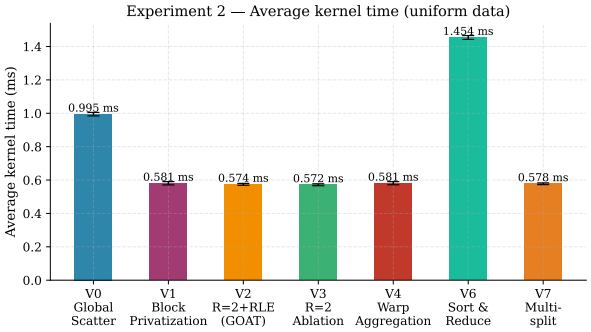
Run	V0 (ms)	V1 (ms)	V2 (ms)	V3 (ms)
1	0.9906	0.5707	0.5683	0.5654
2	0.9986	0.5744	0.5620	0.5766
3	0.9867	0.5714	0.5756	0.5764
4	0.9933	0.5980	0.5792	0.5731
5	0.9936	0.5834	0.5796	0.5647
6	0.9910	0.5732	0.5766	0.5675
7	0.9938	0.5883	0.5796	0.5636
8	0.9828	0.5904	0.5751	0.5789
9	0.9947	0.5704	0.5766	0.5770
10	1.0215	0.5865	0.5712	0.5741
Avg	**0.9947**	**0.5806**	**0.5744**	**0.5717**
Stdev	**0.0104**	**0.0099**	**0.0057**	**0.0058**

Table 4: Kernel execution time — Experiment 2, V4–V7 (uniform, 500k elements, 4096 bins, clipped at 127)

Run	V4 (ms)	V5 (ms)	V6 (ms)	V7 (ms)
1	0.5870	11.9925	1.4661	0.5701
2	0.5705	11.9927	1.4453	0.5728
3	0.5875	11.9839	1.4623	0.5798
4	0.6000	11.9909	1.4454	0.5726
5	0.5797	12.0007	1.4364	0.5838
6	0.5735	11.9852	1.4526	0.5787
7	0.5805	11.9885	1.4691	0.5834
8	0.5805	12.0040	1.4416	0.5838
9	0.5822	11.9900	1.4598	0.5739
10	0.5688	11.9926	1.4632	0.5785
Avg	**0.5810**	**11.9921**	**1.4542**	**0.5778**
Stdev	**0.0092**	**0.0062**	**0.0114**	**0.0051**



(a) Run-by-run timing



(b) Average with std dev

Figure 3: Experiment 2 — kernel execution time, uniform data

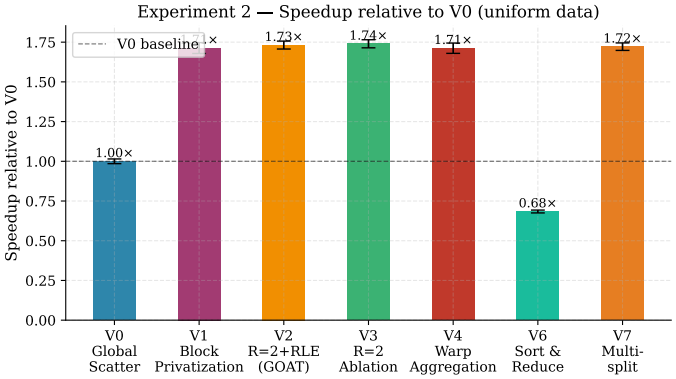


Figure 4: Experiment 2 — speedup of V1, V2, V3 relative to V0 baseline

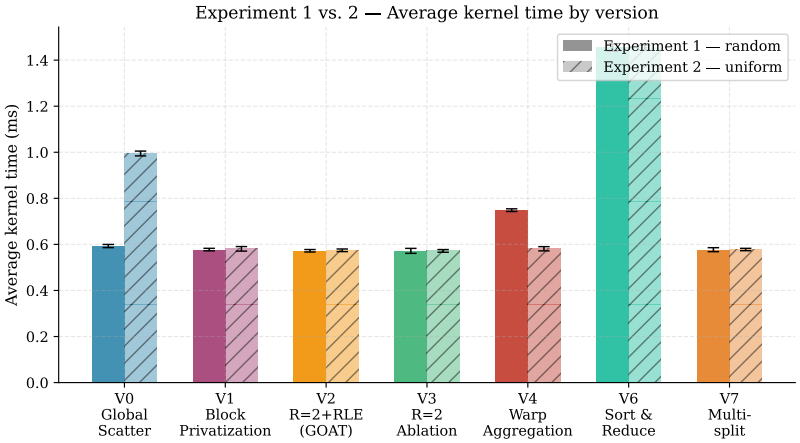


Figure 5: Side-by-side comparison of average kernel time: Experiment 1 vs. 2 per version

bins so each bin receives ≈ 122 hits on average and per-bin contention is low. Compute speedup = 0.5929/0.5774 and discuss here. Reference Figure 2.]

V0 vs. V1 on Experiment 2 (uniform data)

[TODO: Compare 0.9947 ms (V0) vs. 0.5806 ms (V1). All elements target bin 42. V0 serialises every global atomic on one counter. V1 serialises within each block's shared copy then merges once per block to global — far fewer global collisions. Expected: V1 advantage over V0 is *larger* in Experiment 2 than Experiment 1. Reference Figure 4.]

Impact of input distribution on V0

[TODO: Compare 0.5929 ms (Exp1) vs. 0.9947 ms (Exp2) for V0 only. Random data spreads global atomic pressure across 4096 bins; uniform data funnels all threads to one bin, maximally serialising the atomic queue. Reference Figure 5.]

Impact of input distribution on V1

[TODO: Compare 0.5774 ms (Exp1) vs. 0.5806 ms (Exp2) for V1 only. Shared atomics are cheaper so the hot-bin penalty is smaller than in V0. Expect Exp2 to be only modestly slower than Exp1 for V1, in contrast to the larger delta seen in V0. Discuss the ratio.]

4 Analysis 2: Version 2 — Vectorized $\mathcal{R}$-per-Block Privatization

Implementation approach

Version 2 combines *$\mathcal{R}$-per-Block Histogram Replication* ($\mathcal{R} = 2$) with *Vectorized 128-bit Loads*. Two copies of the shared-memory histogram (`multi_histo[0]` and `multi_histo[1]`) are allocated, each padded by one element (4097 `unsigned int`s per copy) to break the 32-bank stride alignment that arises when `num_bins` is a power of two. Even-indexed threads scatter into copy 0; odd-indexed threads scatter into copy 1, so every 32-thread warp splits its atomic traffic evenly across two physical bank regions. After the accumulation phase, a single `__syncthreads()` separates the per-block merge: each thread sums its two copies and issues *one* global `atomicAdd` only when the total is non-zero (sparsity filter). Simultaneously, V2 casts the 32-bit input array to `uint4`, loading four integers per 128-bit global memory transaction and quadrupling effective bandwidth over V0/V1's scalar 32-bit reads.

Superiority over V0 and V1

V0 issues N *global* atomic operations at DRAM latency. V1 migrates these to low-latency shared memory but uses a single histogram copy: in a 32-wide warp, if k threads map to the same 32-wide bank group, those k atomics are serialised in hardware. V2 halves that contention by distributing 32 warp lanes across two buffer copies, reducing the maximum same-bank collision count from 32 (V1) to 16. The +1 padding ensures the two arrays start on different bank offsets, preventing the second copy from merely replicating the same conflict pattern. For the global flush, the sparsity filter eliminates all-zero bins entirely, reducing PCIe/L2 pressure at the per-block merge step.

Performance vs. data distribution

- **Random (Experiment 1):** Bank conflicts are already rare when 4096 bins are addressed randomly; both the `uint4` bandwidth gain and the sparsity filter contribute. See Figure 2.
- **Uniform (Experiment 2):** All 32 warp lanes target the same bin. In V1 all 32 hit the same bank — 32-way serialisation. In V2 they split: 16 go to `multi_histo[0]` and 16 to `multi_histo[1]`, immediately halving the hardware serialisation penalty. The sparsity filter then ensures exactly *one* global atomic fires at the end, regardless of how many thousands of identical elements the block processed. See Figure 4.
- **Static shared memory footprint:** $2 \times 4097 \times 4\,\mathrm{B} = 32{,}776\,\mathrm{B}$, well within P100's 64 KB shared memory limit per SM.

[TODO: After data collection, replace the bullet descriptions above with measured speedup numbers from Tables 1 and 3 and Figures 2–5.]

5 Analysis 3: Version 3 — Ablation: R=2 Without RLE

Implementation approach

Version 3 is a controlled ablation of Version 2. It retains the full $\mathcal{R} = 2$ per-block replication structure, the +1-padded static shared memory (`multi_histo[2][4097]`), the `uint4` 128-bit vectorized load path, and the sparsity-filtered global atomic flush — but removes the register-level RLE state machine. Instead of accumulating consecutive identical bins in a two-register `(current_bin, run_count)` pair, V3 issues an `atomicAdd` to shared memory immediately for every valid element.

Purpose: isolating the RLE contribution

By holding all other algorithmic choices constant (two copies, padding, `uint4`, sparsity filter) and removing only RLE, any measured performance difference between V2 and V3 is attributable *solely* to RLE temporal compression.

Expected behaviour

- **Random (Experiment 1):** Elements are i.i.d. uniform over $[0, 4095]$, so consecutive identical values are rare (probability $\frac{1}{4096}$). RLE run lengths average just above one; V2's state machine fires almost every element anyway. V3 and V2 should post nearly identical performance on random data.
- **Uniform (Experiment 2):** All 500,000 elements are identical. V2's RLE accumulates the entire block's work into a single register counter, issuing *one* shared `atomicAdd` per thread lane per block rather than one per element — dramatically reducing shared memory atomic traffic. V3 fires 500,000 atomics per thread into shared memory, incurring maximum serialisation. The V2 vs. V3 gap should be large for uniform data.

[TODO: After data collection, report exact times from Tables 1 and 3 and discuss the measured V2 vs. V3 delta for each experiment, referencing Figure 6.]

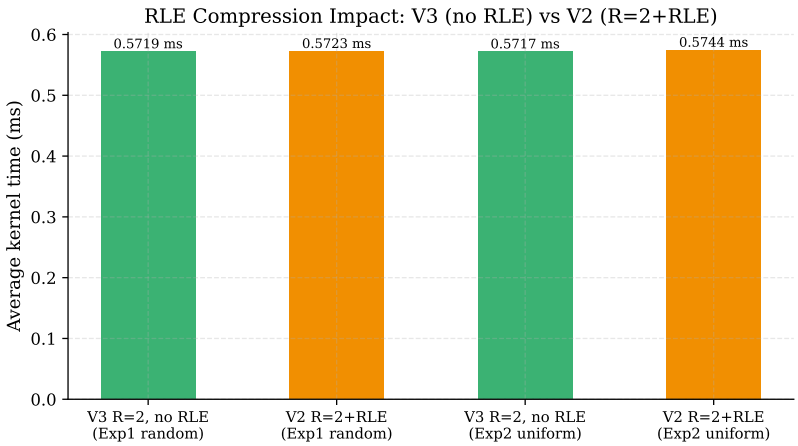


Figure 6: RLE ablation: V3 (R=2, no RLE) vs. V2 (R=2+RLE) for both experiments. Difference isolates the pure temporal compression gain.

6 Additional Kernel Variants

V4 — Warp Aggregation

Version 4 introduces intra-warp aggregation using CUDA warp-level primitives. For each input element, the thread computes its target bin and uses `__ballot_sync` to identify which other lanes in the same warp map to the same bin. One leader (lowest set bit via `__ffs`) reads the group count from `__popc` and issues *one* `atomicAdd` with the aggregated value, while the remaining lanes in the group skip their atomic entirely. This reduces the number of shared-memory atomic operations by a factor equal to the average intra-warp bin collision rate. V4 uses `extern __shared__ unsigned int histo_private[]` sized to `num_bins` 32-bit integers at launch time. The sparsity-filtered global flush is retained from V2.

Expected: On uniform data (Experiment 2), all 32 lanes in a warp target the same bin every iteration, so the aggregation factor is 32 — a full warp reduces to one atomic. On random data, the average collision rate is $32/4096 \approx 0.0078$, so the benefit is modest.

Table 5: V4 (warp aggregation) vs. V1/V2 — average execution time

	Experiment 1 (random)		**Experiment 2 (uniform)**	
Version	**Avg (ms)**	**Stdev**	**Avg (ms)**	**Stdev**
V1 — Block Privatized	0.5774	0.0055	0.5806	0.0099
V2 — R=2+RLE (GOAT)	0.5723	0.0054	0.5744	0.0057
V4 — Warp Aggregation	0.7486	0.0058	0.5810	0.0092

V5 — Bin-Centric Gather

Version 5 inverts the scatter pattern used by V0–V4. Instead of each thread owning an *input element* and scattering it to a bin, each thread owns a *bin* and gathers through the input to count matches. With `num_bins`=4096 and `blockDim`=256, exactly 16 blocks are launched and every bin is covered by exactly one thread. Because no two threads write to the same output location,

zero atomic operations are required anywhere. Shared-memory tiling amortises global memory latency: each block loads a 256-element tile in one coalesced 1024-byte transaction, and all 256 threads scan the tile simultaneously in registers.

Trade-off: The elimination of atomics is achieved at the cost of $O(N \times B)$ comparisons where $B = 4096$. For $N = 500{,}000$ this means 2×10^9 comparison iterations. Performance is arithmetic-bound, not memory-bandwidth-bound, and is dominated by loop count regardless of data distribution.

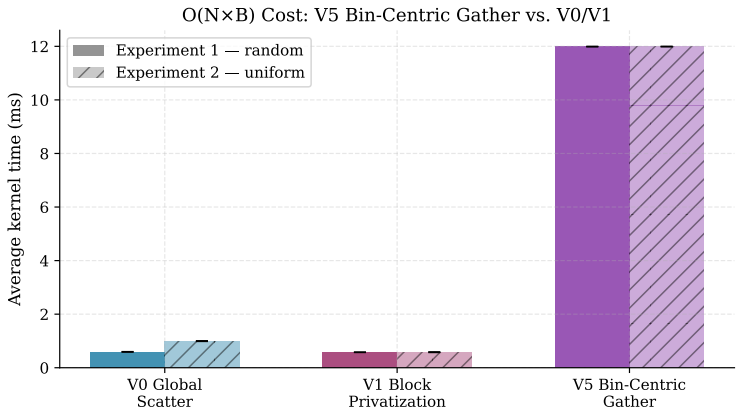


Figure 7: V5 bin-centric gather vs. V0/V1: $O(N \times B)$ arithmetic cost vs. atomic-based versions. V5 is much slower in absolute time despite issuing zero atomics.

V6 — Sort & Reduce-by-Key via Thrust

Version 6 reframes histogram construction as a sort problem. The device input array is radix-sorted in-place via `thrust::sort`, converting the unordered element stream into a contiguous sequence of equal values. A subsequent `thrust::reduce_by_key` collapses each contiguous run to a (bin, count) pair in a single pass. `thrust::scatter` then places the counts at their correct positions in the output bin array, and a final `convert_kernel` applies the 127-cap clipping.

This approach requires no atomic operations and avoids shared-memory contention entirely. However, it incurs an $O(N \log N)$ GPU radix sort (Thrust uses CUB internally) before the $O(N)$ reduction. Because `thrust::sort` modifies `d_input` in-place, each experimental run reloads input from disk to preserve correctness.

Expected: Sort latency dominates. Likely slower than V1/V2 for random data, but may be competitive with V5 (Gather) for highly skewed data where the sorted stream is nearly run-length-compressed already.

V7 — Multi-split Hierarchical Bucketing

Version 7 applies a two-level hierarchical decomposition of the 4096-bin histogram. Each 12-bit bin index is split as super_bucket $= b \,/\, 256$ and offset $= b \bmod 256$, with the shared-memory target address computed as target $=$ super_bucket $\times 256 +$ offset (which equals b, preserving correctness while encoding a locality hint in the access pattern).

With 16 super-buckets of 256 bins each, threads in a warp that hash to the same super-bucket compete for nearby shared-memory addresses within one 256-element region, improving cache-line reuse and reducing the probability of cross-region bank conflicts compared to a flat 4096-bin address

space. V7 uses `extern __shared__ unsigned int histo_private[]` (same dynamic size as V4) and the sparsity-filtered global flush.

Table 6: V4, V6, and V7 — average execution time summary

	Experiment 1 (random)		**Experiment 2 (uniform)**	
Version	**Avg (ms)**	**Stdev**	**Avg (ms)**	**Stdev**
V4 — Warp Aggregation	0.7486	0.0058	0.5810	0.0092
V6 — Sort & Reduce	1.4545	0.0156	1.4542	0.0114
V7 — Multi-split	0.5770	0.0087	0.5778	0.0051

7 Short Answer Questions

Global memory reads per kernel

All scatter-based kernels (V0–V4, V7) read the input array exactly once — each of the N elements is loaded by exactly one thread. V5 (Bin-Centric Gather) effectively reads the entire input for each of B bins, yielding $O(N \times B)$ global load operations. V6 reads the input once for `thrust::sort`, which then operates in-place. **Global reads: N for V0, V1, V2, V3, V4, V6, V7; $O(N \times B)$ for V5.**

Global memory writes per kernel

- **V0:** Every `atomicAdd(&bins[b], 1)` is a global read-modify-write. N **global atomic writes total.**
- **V1, V2, V3, V4, V7:** Accumulation uses shared memory. The merge phase issues at most BLOCKS $\times K$ global atomic writes, where $K \leq 4096$ is the number of non-zero bins. For random data $K \approx 4096$; for uniform data $K = 1$.
- **V5:** One non-atomic write per bin: exactly $B = 4096$ global writes total, regardless of N or distribution.
- **V6:** `thrust::sort` operates in-place on *d_input*; `thrust::scatter` writes at most $N_{\text{unique}} \leq B$ locations to the bin array.

Atomic operations per kernel

- **V0:** N global atomics — one per element.
- **V1:** N shared atomics (accumulation) + up to BLOCKS $\times K$ global atomics (merge). Total: $N + \text{BLOCKS} \cdot K$.
- **V2:** N shared atomics split across $\mathcal{R}$=2 copies (RLE reduces the effective count for long runs) + up to BLOCKS $\times K$ global atomics (sparsity-filtered merge).
- **V3:** Same structure as V2 without the RLE reduction step — all N shared atomics fire unconditionally.
- **V4:** N/agg shared atomics (warp-aggregated, where agg is the average same-bin collision count per warp) + up to BLOCKS $\times K$ global atomics (sparsity-filtered merge).
- **V5: Zero** — no `atomicAdd` anywhere. Each bin is owned by exactly one thread; no concurrent writes occur.
- **V6: Zero** post-sort — Thrust radix sort and reduce-by-key use their own internal synchronisation primitives, not user-level atomic operations.

- **V7:** N shared atomics (same count as V1; locality hint does not reduce the number of atomics, only their distribution within the shared-memory address space) + up to BLOCKS $\times K$ global atomics (sparsity-filtered merge).